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Regularized Market Timing with Transaction Costs

Evidence from Daily SPY Returns

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Member	Full name	Student ID
1	Li Chongkai	A0357831X
2	Zhang Leyao	A0356596J
3	Zhang Yikai	A0355997B
4	Ye Tiantian	A0354470E

Abstract

We investigate whether a small return-prediction model improves daily equity timing after transaction costs. Five trailing trend and volatility features are used to forecast SPY opening-to-opening returns. Ridge regression is refitted monthly on the latest 1,260 realized outcomes. Four regularization strengths are compared during 2019–2021, and the selected rule is evaluated over 2022–2025. Positive forecasts allocate the portfolio to SPY; other forecasts allocate it to cash. At a single-side cost of 5 basis points, the strategy earns 6.54% compound annual growth and a 0.481 Sharpe ratio, compared with 11.18% and 0.670 for buy-and-hold. Its lower drawdown accompanies 69.36% average equity exposure. The forecast's out-of-sample R^2 is -0.00541 relative to a past-only historical mean. Gross Sharpe is close to the passive benchmark, but 246 charged trade units impose 3.09% annualized log-wealth drag. The results connect weak prediction, binary position changes, and trading costs: the regularized estimation procedure is reproducible, while the resulting timing rule does not improve net investment performance.

Data: January 2010–December 2025. Final evaluation: January 2022–December 2025.

1 Introduction

The investment value of a return forecast depends on the positions it produces and the cost of changing those positions. A model with a small statistical signal can generate frequent trades, while a strategy with lower volatility can simply spend less time invested. We study these issues through a daily SPY timing strategy whose inputs, estimation, executions, and accounting are specified together.

The research question is whether recent price trends and volatility, combined through a regularized linear model, improve the allocation between SPY and cash. The feature set consists of returns over one, five, and twenty sessions, the distance from a twenty-session moving average, and recent return volatility. This specification allows short-term continuation and reversal to compete in estimation. Trends can persist when information and portfolio adjustments are gradual; temporary price pressure can instead reverse. Volatility supplies an observable measure of the environment in which those effects occur.

Ridge regression is a suitable estimator for this design because several inputs summarize overlapping price history. Its ℓ_2 penalty stabilizes their joint coefficients while retaining all five predictors [1]. The strategy is updated monthly using a fixed-length rolling window. A small validation search selects the penalty, and a separate historical period evaluates the selected trading rule. This connects model fitting to an investor's decision rather than treating forecast accuracy as the sole objective.

The distinction between statistical and economic performance is central in financial prediction. Chinco et al. [2] combine regularized forecasts with an explicit trading rule at an intraday horizon; Kelly et al. [3] examine how shrinkage and model complexity affect return forecasts and timing portfolios. Our study applies these design principles to a small daily model. A rolling historical mean provides a forecast reference, following the benchmark discipline emphasized by Campbell and Thompson [4]. SPY buy-and-hold and a fixed moving-average strategy provide the investment comparisons.

Three questions organize the analysis. Does the model improve prediction relative to the historical mean? Does the selected trading rule improve performance after costs? How much of its lower risk reflects reduced equity exposure? Answering them requires a consistent chain from the information available at the decision close to the return earned after the next opening execution.

The final evaluation covers 1,002 holding intervals in 2022–2025. At 5 basis points per one-way trade, Ridge timing earns 6.54% CAGR and a 0.481 Sharpe ratio, below buy-and-hold's 11.18% and 0.670. Its maximum drawdown is smaller, but it is invested in SPY during only 69.36% of intervals. Prediction error is slightly higher than the historical-mean benchmark. The economic explanation is therefore not simply that costs destroy a strong forecast: gross risk-adjusted performance is already close to the passive alternative, and frequent full-position switches then reduce net returns.

The report follows that explanation. Sections 2–4 establish the information set, estimation procedure, and portfolio accounting. Section 5 links prediction diagnostics to wealth, costs, exposure, and annual outcomes. The final sections discuss the investment implications, limitations, and reproducibility of the study.

2 Data and Executable Returns

2.1 Market, source, and data quality

SPY is a US-listed exchange-traded fund tracking the S&P 500 and serves as both the traded asset and the passive benchmark [5]. All positions and returns are denominated in US dollars. Daily observations are indexed by New York exchange-session dates, including shortened sessions. No currency conversion is required.

Historical prices are downloaded from Yahoo Finance through `yfinance`, with `auto_adjust=False`, corporate actions enabled, and automatic repairs disabled [6]. The data contain separate open, high, low, close, adjusted close, volume, dividend, and split fields. Adjusted close is used to construct trailing features; unadjusted opening prices and explicit distributions determine portfolio returns.

Table 1. Historical data and quality checks.

Item	Recorded result
Requested history	1 January 2010 to 1 January 2026, end date excluded
Observed sessions	4 January 2010–31 December 2025; 4,024 sessions
Trading-calendar check	4,024 expected XNYS sessions; no missing or extra dates
Duplicate dates / missing values	0 / 0
Dividend / split events	64 / 0
Zero-volume sessions	0
Feature warm-up / terminal exclusions	20 initial sessions / 2 unavailable forward outcomes
Complete labeled observations	4,002 across the full history

Source: `output/data_audit.json`. XNYS session dates are verified with `exchange_calendars`.

The cleaning routine sorts dates, verifies unique observations, and checks finite positive prices, nonnegative distributions and volume, and OHLC ordering. Expected sessions are checked against the exchange calendar, including special closures [7]. Weekends and holidays are not filled. Missing vendor fields and unsupported splits trigger an error. Observed extreme returns are retained. These rules leave the sample in Table 1; no full-sample outlier threshold is estimated.

The source snapshot records its retrieval time, download options, package version, and SHA-256 digest. This identifies the exact input used by the published experiment. Representative distribution checks and the provenance files are summarized in Appendix A.

2.2 Decision, execution, and label maturity

Let t index trading sessions. Features and the forecast are calculated after close t. The resulting target position is executed at open $t + 1$ and held until open $t + 2$. With unadjusted opening price O_t and distribution D_t whose ex-date is t, the target is

$$y_t = \frac{O_{t+2} + D_{t+2}}{O_{t+1}} - 1. \tag{1}$$

A share held into the exit ex-date earns that distribution. A share first purchased at the ex-date open does not earn the entry-date distribution. The backtest credits the entitled dividend receivable at exit and applies the same convention to all strategies.

Table 2. Information timing for the first test observation.

Event	Session	Information used or realized
Signal calculation	31 December 2021 close	Trailing features and the fitted model
Position execution	3 January 2022 open	Target position from the previous close
Holding-period exit	4 January 2022 open	Opening-to-opening return and any entitled dividend
Training eligibility	By 4 January 2022 close	The complete outcome is now observed

Label maturity determines which rows can enter training. At decision close d, a return ending at that morning's open is observed, whereas a return ending at the next session's open is not. Writing $e(t) = t+2$ for the exit session, the training window is

$$\mathcal{T}_d = \text{tail}_{1260}\{t : t < d,\ e(t) \le d\}. \tag{2}$$

The timestamp condition is applied to realized outcomes as well as to feature dates. The latest eligible signal can therefore be more than one session behind the current decision. Every fit exports its first and last training signal, maximum label exit, and sample count.

2.3 Features and their economic motivation

Let A_t be adjusted close and $r_t^A = A_t/A_{t-1} - 1$. Table 3 defines the complete specification. All five features are known after close t.

Table 3. The five fixed predictors.

Feature	Definition	Predictive role
One-session return	$A_t/A_{t-1} - 1$	Recent continuation or reversal
Five-session return	$A_t/A_{t-5} - 1$	Approximately weekly trend
Twenty-session return	$A_t/A_{t-20} - 1$	Approximately monthly trend
Moving-average gap	$A_t/\left(\frac{1}{20}\sum_{j=0}^{19} A_{t-j}\right) - 1$	Price location relative to its recent mean
Return volatility	Sample SD of $r_{t-19}^A, \ldots, r_t^A$	Recent return uncertainty

The volatility uses 20 returns and denominator 19. The longest required price lookback is 20 sessions. Implementation: `project2/core.py::make_samples`.

The three return horizons give the model several views of recent price movement. The moving-average gap captures the cumulative location of the price path, while volatility contributes information about the scale of recent movements. Because these predictors share price history, their effects are estimated jointly. A later common multiplicative adjustment of the historical price window cancels from the ratios and return volatility; Appendix A gives the corresponding algebra.

The horizons distinguish immediate price movement from approximately weekly and monthly behavior. A positive return coefficient is consistent with continuation over that feature's horizon; a negative coefficient represents reversal after conditioning on the other inputs. Neither sign is imposed during fitting. The moving-average gap adds information about the location of the endpoint relative to the intervening path: two windows can have the same twenty-session return and different average price levels. Volatility adds a scale measure that the signed return features do not supply. These choices express a small set of competing economic mechanisms, rather than assuming that all recent gains must predict further gains.

SPY makes the allocation question concrete. The same fund is used for signal construction, execution, and the passive investment comparison, avoiding an additional stock-selection problem. Restricting the study to one traded asset also keeps the accounting focused on the forecast-to-position decision: any difference from buy-and-hold comes from when capital is invested and the cost of changing that allocation. The data source supplies both adjusted prices for trailing signals and the raw prices and distributions needed to make that distinction explicit.

3 Model, Training, and Validation

3.1 Why a small regularized linear model?

The specification asks whether a compact summary of recent price behavior contains a useful conditional-mean signal. A linear model estimates the joint contribution of the five inputs without expanding the search into transformations and interactions. This is a substantive restriction: the expected return is represented as an additive function of recent returns, price location, and volatility. The volatility coefficient changes the expected-return forecast; it does not mechanically scale the position by inverse volatility.

The main estimation concern is the combination of noisy daily outcomes and overlapping predictors. Five- and twenty-session returns share observations, and the moving-average gap also summarizes the recent price path. Ordinary least squares can allocate large, offsetting coefficients to similar inputs. Ridge penalizes the squared size of that coefficient vector, favoring a more stable joint forecast. Here there are five inputs and 1,260 training observations, so shrinkage addresses estimation variability rather than a shortage of observations relative to predictors.

Retaining all five inputs also matches the research premise. Each represents a related but distinct summary of recent behavior, and the design does not assume that only one is relevant. Ridge therefore provides a natural alternative to a sparsity penalty such as LASSO, which can remove predictors entirely. A larger nonlinear model would require additional choices about interactions, capacity, and tuning. The implemented experiment instead concentrates its model search on one transparent source of flexibility, the shrinkage strength. These are design reasons for the chosen estimator; the empirical comparison is against the recorded forecast and trading benchmarks.

3.2 Ridge estimation on past-only standardized inputs

At each refit date d, means μ_d and population standard deviations s_d are estimated on $\mathcal{T}_d$. For each feature coordinate, the standardized value is

$$z_{t,d} = (x_t - \mu_d) \oslash s_d, \tag{3}$$

where $\oslash$ denotes coordinatewise division. The same transformation is used for subsequent predictions until the next refit. The model minimizes

$$(\widehat{a}_d, \widehat{\beta}_d) = \arg\min_{a,\beta} \left\{ \sum_{t \in \mathcal{T}_d} (y_t - a - z_{t,d}^{\mathsf{T}}\beta)^2 + \alpha \|\beta\|_2^2 \right\}. \tag{4}$$

The intercept is unpenalized. The implemented loss is a sum of squares, so α has the scikit-learn Ridge convention [8].

The penalty shrinks coefficients that would otherwise be sensitive to overlapping return predictors. For centered training data with singular-value decomposition $Z = USV^{\mathsf{T}}$, the solution can be written as

$$\widehat{\beta}_d = V \operatorname{diag}\left(\frac{s_j}{s_j^2 + \alpha} \right) U^{\mathsf{T}} y_c. \tag{5}$$

This replaces the least-squares amplification $1/s_j$ by $s_j/(s_j^2 + \alpha)$, reducing sensitivity in directions with small singular values. Standardization makes the penalty comparable across features measured on different scales. The implementation uses a `StandardScaler`/`Ridge` pipeline with the SVD solver. It does not use stochastic gradient training.

The unpenalized intercept gives the historical-mean benchmark a direct interpretation. Since the standardized training columns have zero mean, minimizing Eq. (4) over the intercept yields

$$\widehat{a}_d = \overline{y}_d, \qquad \widehat{y}_t = \overline{y}_{d(t)} + z_{t,d(t)}^{\mathsf{T}} \widehat{\beta}_{d(t)}. \tag{6}$$

Ridge thus adds a conditional adjustment to the same past-only mean used by the prediction reference. Increasing α shrinks the adjustment toward zero; in the limit the forecast approaches that reference. This separates an estimated average equity return from the incremental information attributed to the five features.

The model is fitted on the first evaluation signal and the first decision session of each new calendar month. Each fit uses the latest 1,260 mature labels, with a minimum requirement of 1,000. Between fits, the coefficients and scaler remain fixed, while new daily features change the forecasts. If $d(t)$ denotes the latest refit date, then

$$\widehat{y}_t = \widehat{a}_{d(t)} + z_{t,d(t)}^{\mathsf{T}} \widehat{\beta}_{d(t)}. \tag{7}$$

The five-year window balances a substantial estimation sample against the use of a full expanding history whose oldest observations never leave the fit. Monthly updates allow the estimated relationship to change on a regular schedule while avoiding daily coefficient re-estimation. Daily decisions still respond to new features between refits. The window and update frequency are fixed parts of the specification, so the validation search compares four penalties under the same adaptation rule rather than jointly searching over many training designs.

3.3 Chronological split and forward validation

The initial development history is 2010–2018. Holding intervals within 2019–2021 form the validation period, and intervals within 2022–2025 form the final evaluation. Period membership is defined by both entry and exit dates. This gives the test entry on 3 January 2022 and liquidation at the 31 December 2025 open.

Forward validation reproduces the intended use of the model: fit with outcomes already realized, issue a forecast, and move to the next decision. Randomly shuffled folds would allow later observations to inform forecasts for earlier dates and would break that deployment sequence. The label-exit filter in Eq. (2) handles the additional execution delay explicitly. Adjacent holding intervals share an opening-price boundary but do not share the same holding return; eligibility is determined by when the entire label is observed, rather than by an arbitrary gap between rows.

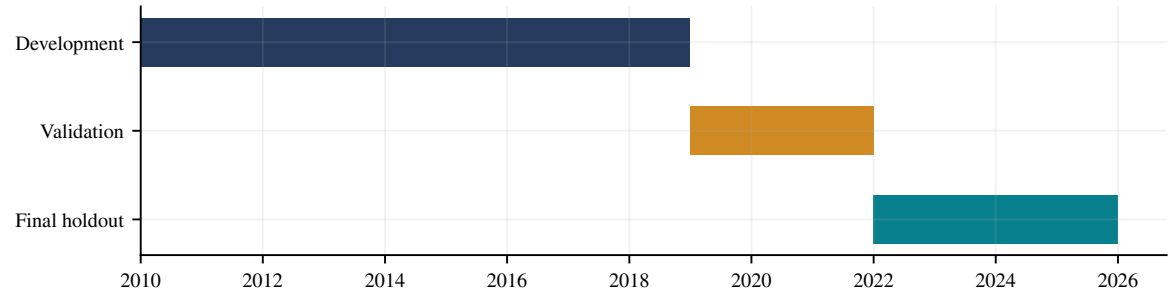


Figure 1. Chronology of the research design. All fits use the latest 1,260 fully realized labels. The penalty is selected using validation performance and then held fixed during final evaluation.

Four penalty values, $\alpha \in \{0.1, 1, 10, 100\}$, are walked forward through validation. Each candidate has its own monthly past-only fits and daily position path. The selection criterion is net validation Sharpe at a single-side cost of 5 basis points. Ties within 10^{-12} are resolved in favor of the larger penalty. Table 4 reports every candidate.

Penalties 0.1 and 1 produce identical validation position signs and hence identical trading results. Their continuous forecasts differ, but the tie rule selects $\alpha = 1$. Larger penalties have lower validation Sharpe. Forecast MSE is calculated as a diagnostic; it is not an additional selection objective. This distinction

Table 4. Forward validation, 2019–2021: the complete four-candidate search.

α	CAGR (%)	Net Sharpe	Exposure (%)	Turnover	R^2_{OOS}
0.1	22.33	1.2060	76.72	136	-0.01442
1	22.33	1.2060	76.72	136	-0.01434
10	19.15	1.0767	77.38	134	-0.01364
100	18.64	1.0310	85.71	116	-0.01089

Each candidate is evaluated on 756 holding intervals at 5 bps. CAGR and exposure are percentages; turnover counts charged one-way trade units. R^2_{OOS} uses the past-only historical-mean reference and is reported as a diagnostic.

matches the research question: estimation minimizes squared prediction error, while validation selects the regularization strength for the trading application.

The grid illustrates why those objectives differ. At $\alpha = 100$, validation R^2_{OOS} improves from approximately -0.01434 to -0.01089, yet net Sharpe falls from 1.2060 to 1.0310. The sign rule depends on which scores cross zero, not on their squared errors alone. Exposure rises from 76.72% to 85.71%, while turnover falls from 136 to 116 units. Stronger shrinkage therefore changes both the held-return subset and its implementation costs. The stated Sharpe rule resolves that trade-off without selecting a different criterion after observing the final period.

3.4 Final rolling evaluation and overfitting control

The selected penalty is used throughout final evaluation. Earlier test outcomes enter later training windows once their exit dates have passed. The model therefore adapts through a fixed rolling rule rather than remaining fitted to a single pre-2022 sample. The final path contains 49 fits: an initial fit at the 31 December 2021 close, followed by the first decision session in each month of 2022–2025. Every test fit contains 1,260 eligible observations.

The search is restricted to four penalties for one asset, one feature set, one training-window length, and one monthly update schedule. The forecast threshold and moving-average benchmark are fixed. These choices control the number of alternatives considered, while Ridge controls coefficient size within each alternative. Training-only scaling and the label-maturity filter separately control information leakage.

The procedure is implemented in four stages:

1. Construct trailing features and the opening-to-opening target; retain fully defined observations.
2. Walk each penalty through validation, using Eq. (2) at every refit, and account for the resulting trades at the declared cost.
3. Select α using the stated Sharpe and tie rules; record the protocol, input, and validation-table fingerprints.
4. Generate the selected model's test forecasts and positions, then save daily accounting paths, monthly fit audits, and evaluation tables.

The exported audit records make the correspondence between training and testing explicit. Unit checks also change future prices and verify that earlier features, fits, scalers, and forecasts remain unchanged. Source functions and reproduction commands are listed in Appendix C.

4 Portfolio Construction and Evaluation

4.1 Forecast-to-position rule and investment benchmarks

The portfolio follows the fixed sign rule

$$w_t = \mathbb{I}\{\widehat{y}_t > 0\}. \tag{8}$$

A positive prediction allocates all capital to SPY at the next open; otherwise the portfolio holds cash. Positions are unlevered and long-only, with a maximum equity weight of one. Cash earns zero, and the Sharpe reference rate is also zero.

The zero threshold follows the gross allocation comparison under these constraints: a positive predicted SPY return favors equity over zero-return cash. The binary map keeps that directional decision separate from an additional position-sizing model. It also makes each threshold crossing economically consequential, because the entire portfolio switches assets. Transaction costs enter validation and all net performance calculations; they do not shift the trading threshold or introduce a no-trade region in the implemented rule.

Two investable strategies use the same holding dates. Buy-and-hold remains in SPY throughout. The fixed MA20 strategy holds SPY when adjusted close exceeds its trailing twenty-session average, equivalently when the moving-average gap in Table 3 is positive. It uses the same next-open execution as Ridge. Neither its window nor its threshold is selected in validation.

The two comparisons have different roles. Buy-and-hold measures whether active allocation improves on retaining the traded market exposure. MA20 asks whether estimating a combination of signals improves on a direct, fixed trend rule built from the same information set. Giving all strategies the same opening-to-opening returns, distributions, and fee convention makes their differences attributable to their positions and turnover rather than to mismatched price series or evaluation dates.

Prediction performance is compared with the historical mean of the same mature training outcomes used at each monthly refit. This reference forecast remains fixed between refits, matching the main model's update schedule. It provides a simple alternative against which to assess squared-error prediction.

4.2 Transaction costs and compounded wealth

Index the evaluated holding intervals by $i = 1, \ldots, N$. The portfolio starts from cash, $w_0 = 0$, and the one-way turnover at entry is $q_i = |w_i - w_{i-1}|$. With single-side fee c, its interval growth factor is

$$G_i(c) = (1 - cq_i)(1 + w_i y_i). \tag{9}$$

Fees are deducted as a fraction of pre-trade equity before earning the interval return. Terminal liquidation adds a factor $1 - cw_N$. Thus

$$V_N(c) = \left[\prod_{i=1}^{N} G_i(c)\right](1 - cw_N), \qquad V_0 = 1. \tag{10}$$

An unchanged SPY or cash position has no trading charge. Buy-and-hold pays once on entry and once on liquidation. Binary positions require no intermediate fractional-weight rebalancing.

The main cost is $c = 0.0005$, or 5 basis points per one-way unit. Zero- and 10-basis-point scenarios use the same predictions and positions. This isolates the effect of fees on the frozen trading path. Calendar-year returns are extracted from the continuous portfolio path; year boundaries do not create additional transactions.

The cost mechanism has a simple exact form. With $K = \sum_i q_i + w_N$, every charged trade multiplies wealth by $1 - c$, so

$$\frac{V_N(c)}{V_N(0)} = (1-c)^K. \tag{11}$$

The annualized log-wealth drag is $-(252K/N)\log(1-c)$. This quantity is additive in log wealth and is distinct from the percentage-point difference between two CAGRs. The numerical attribution is reported with the cost results in Section 5.2.

4.3 Performance measures

Let R_i be the net interval return after all applicable fees, including final liquidation. We report the metrics in Table 5. Annualization uses 252 holding intervals. The reported sample ends at the final opening liquidation rather than the final day's close.

Table 5. Evaluation definitions.

Measure	Definition
Total return	$V_N - 1$
Compound annual growth	$\text{CAGR} = V_N^{252/N} - 1$
Arithmetic annual return	$252\overline{R}$
Annualized volatility	$\sqrt{252}\ \text{sd}(R_i)$, with sample SD
Sharpe, $r_f = 0$	$252\overline{R}/[\sqrt{252}\,\text{sd}(R_i)]$
Maximum drawdown	$\min_i [V_i/\max(1, V_1, \ldots, V_i) - 1]$
Average equity exposure	$N^{-1}\sum_i w_i$
Total one-way turnover	$K = \sum_i \lvert w_i - w_{i-1}\rvert + w_N$

For the historical-mean prediction $\bar{y}_{d(i)}$, out-of-sample explanatory performance is

$$R^2_{\text{OOS}} = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y}_{d(i)})^2}. \tag{12}$$

The denominator uses a past-only reference, not the test-period mean. Positive values indicate lower squared error than that reference. Directional accuracy and conditional return summaries complement this measure by connecting the sign rule to the realized returns held or missed.

5 Results and Economic Interpretation

5.1 Net performance relative to the investment benchmarks

Table 6 reports final-period results at 5 basis points per one-way trade. Ridge timing earns 6.54% CAGR, compared with 11.18% for buy-and-hold and 3.69% for MA20. Its arithmetic annual return is 7.56%, annualized volatility is 15.72%, and Sharpe is 0.481. The passive strategy has a higher return and Sharpe, while MA20 has the lowest volatility and growth.

The terminal wealth values are 1.286 for Ridge, 1.155 for MA20, and 1.524 for buy-and-hold. Figure 2 shows how the difference develops over the full holding period. Ridge reduces losses during parts of 2022, but subsequently participates in fewer of the positive equity returns. Its lead over MA20 does not translate into an advantage over the passive alternative.

The magnitude of the trade-off is economically important. Ridge's CAGR is 4.64 percentage points below buy-and-hold, while its maximum drawdown is 4.08 percentage points smaller in absolute magnitude. These quantities describe different aspects of the wealth path, so the drawdown improvement does

Table 6. Final evaluation, 2022–2025: performance after 5 bps single-side costs.

Metric	Ridge	MA20	Buy-and-hold
CAGR (%)	6.54	3.69	11.18
Arithmetic return (%)	7.56	4.22	12.27
Volatility (%)	15.72	10.83	18.31
Sharpe ($r_f = 0$)	0.481	0.389	0.670
Max. drawdown (%)	-22.20	-19.14	-26.28
Total return (%)	28.63	15.52	52.38
Equity exposure (%)	69.36	65.27	100.00
One-way turnover	246	102	2

All strategies use the same 1,002 opening-to-opening intervals, initial capital, dividend convention, and liquidation date. Returns and volatility are annualized except for total return. Equity exposure is the proportion of intervals invested.

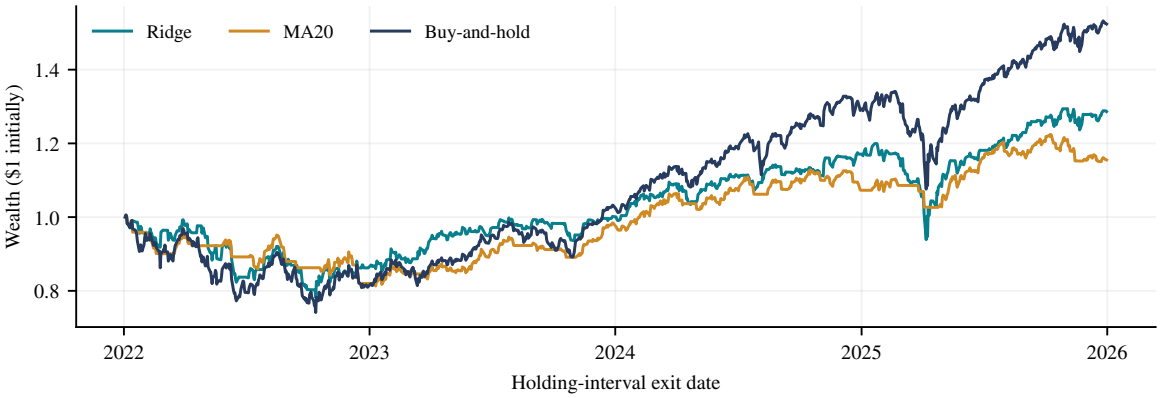


Figure 2. Cumulative net wealth from one unit of initial capital. Entry and final liquidation are included in every path. Source: `output/tables/test_path_*.csv`.

not offset the growth difference automatically. The remaining results examine the prediction, cost, and exposure components separately.

5.2 Transaction costs and the implementation gap

At zero cost, Ridge earns 9.88% CAGR and a Sharpe of 0.678. Buy-and-hold earns 11.20% and 0.672. Thus gross risk-adjusted performance is numerically close, although gross growth is already lower for the timing rule. At 5 bps, Ridge's Sharpe falls to 0.481; at 10 bps it falls to 0.284. Figure 3 and Table 7 isolate this change while keeping the position path fixed.

Table 7. Ridge and buy-and-hold at the three declared cost levels.

bps	Ridge CAGR (%)	Sharpe	BH CAGR (%)	BH Sharpe
0	9.88	0.678	11.20	0.672
5	6.54	0.481	11.18	0.670
10	3.29	0.284	11.15	0.669

CAGR values are percentages; BH denotes buy-and-hold. The complete three-strategy comparison is in Appendix B.

Ridge incurs 246 charged one-way trade units, equivalent to 123 completed entry–exit round trips. Annualized turnover is 61.87 one-way units. MA20 incurs 102 units, and buy-and-hold incurs two. The contrast explains why the passive strategy's cost curve is almost flat while Ridge's declines sharply.

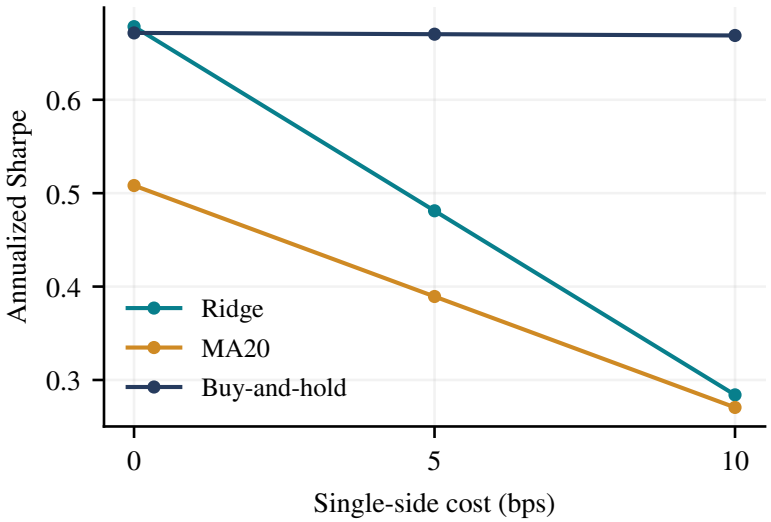


Figure 3. Sharpe sensitivity to the single-side fee. The same forecasts and positions are priced at 0, 5, and 10 bps.

Equation (11) gives the exact wealth reduction from the assumed fee. For Ridge,

$$\frac{V_N(0.0005)}{V_N(0)} = (0.9995)^{246} = 0.884236. \tag{13}$$

The fee removes 11.58% of terminal gross wealth, corresponding to 3.094% annualized log-wealth drag. MA20's corresponding drag is 1.283%, and buy-and-hold's is 0.025% (Table 8). The arithmetic annual return for Ridge falls from 10.66% gross to 7.56% net. These calculations quantify an implementation mechanism that is distinct from a change in the forecasting model.

Table 8. Cost attribution at 5 bps using the exact turnover identity.

Strategy	K	Net/gross wealth	Annual log drag (%)
Ridge	246	0.884236	3.094
MA20	102	0.950267	1.283
Buy-and-hold	2	0.999000	0.025

Annual log drag is $-(252K/N)\log(1-c)$; it is not the percentage-point change in CAGR.

The entry scores help explain why frequent switches are expensive. There are 123 cash-to-SPY entries, with a median predicted holding return of 4.17 bps; 56.10% of entry scores are below 5 bps. Each invested run lasts 5.65 holding intervals on average. A forecast of one interval's return therefore triggers a position whose eventual duration is determined by subsequent signs. The initial score measures neither the complete run's expected return nor its total fee burden.

For scale, a position entered and liquidated after one interval needs an asset return of $(1-c)^{-2}-1$, or approximately 10.01 bps at $c = 0.0005$, just to recover its two trading charges. A multi-interval run spreads those charges over a longer period. This accounting comparison explains why a positive score alone can be insufficient for profitable implementation: the zero threshold responds to the forecast's sign, whereas fees depend on position transitions and eventual exit. The recorded entry scores and holding durations locate that mismatch in the actual trading path.

5.3 Forecast quality and the scale of the trading signal

The model's squared-error performance does not improve on the rolling historical mean. Its MSE is 0.00013365, compared with 0.00013293 for the reference, giving $R^2_{\text{OOS}} = -0.00541$. The prediction

error is approximately 0.54% higher than the baseline error. The forecast–outcome correlation is 0.0295.

Equation (6) allows the incremental forecast to be assessed directly. Let m_i be the historical-mean forecast, $\delta_i = \widehat{y}_i - m_i$ the feature adjustment, and $e_i = y_i - m_i$ the baseline error. The exact sample identity is

$$\mathrm{MSE}(\widehat{y}) - \mathrm{MSE}(m) = \overline{\delta^2} - 2\overline{e\delta}. \tag{14}$$

In squared-basis-point units, the first term is 162.24 and the second is 90.28, leaving an increase of 71.96. The adjustment aligns positively with the baseline error, but its squared magnitude exceeds the error reduction from that alignment. The negative out-of-sample R^2 therefore reflects an adjustment that is too variable relative to its realized predictive contribution, rather than an absence of any positive forecast association.

Table 9. Prediction diagnostics for the frozen final-period forecasts.

Measure	Value
Model MSE	0.00013365
Historical-mean MSE	0.00013293
R^2_{OOS}	-0.00541
Direction accuracy (%)	51.20
Positive-return fraction (%)	53.49
Forecast–outcome correlation	0.0295
Mean absolute forecast (bps)	9.92
Forecasts within 5 bps of zero (%)	37.33

The reference mean uses the same mature training window and monthly update schedule as Ridge. The positive-return fraction is the accuracy of an always-positive direction forecast.

Directional accuracy is 51.20%, below the 53.49% fraction of positive underlying intervals. The model predicts a positive return in 695 of 1,002 cases. Of those invested cases, 371 realize positive returns; among 307 cash cases, 142 realize non-positive returns. The model therefore has no direction-accuracy advantage over an always-positive forecast in this sample. The portfolio comparison additionally weights each outcome by its magnitude and accounts for the trades.

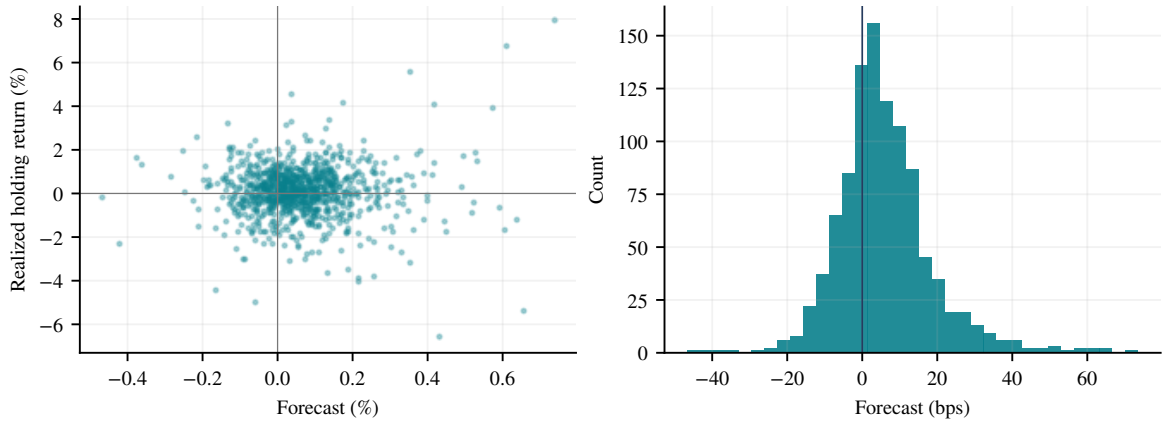


Figure 4. Forecast scale and realized returns. Left: each point is a predicted and realized holding return. Right: the distribution of forecast scores, with the trading threshold at zero.

Figure 4 shows the scale mismatch. The standard deviation of forecasts is 12.80 bps, versus 115.33 bps for realized holding returns. The mean absolute forecast is 9.92 bps, and 37.33% of forecasts are within

5 bps of zero. Under the binary rule, those small positive forecasts receive the same allocation as the largest positive forecasts. When a prediction crosses zero, the entire portfolio changes between SPY and cash. The score distribution therefore helps connect modest predictive variation to large one-way trade amounts.

The returns held and missed provide another useful decomposition. Mean SPY return is 6.097 bps in invested intervals and 2.121 bps in cash intervals (Table 10). Although the invested subset has the higher mean, the excluded subset also has positive average returns. Writing p for the invested fraction, the full asset mean is

$$\overline{y} = p\,\overline{y}_{\text{invested}} + (1-p)\,\overline{y}_{\text{cash}}, \tag{15}$$

whereas the strategy's gross mean is only $p\,\overline{y}_{\text{invested}}$. With $p = 0.6936$, the forgone cash-interval asset returns account for about 1.64 percentage points of arithmetic annual return before fees. Gross Ridge return is therefore 10.66% rather than the passive 12.30%, even though the invested subset has a higher mean return.

Table 10. Realized asset returns grouped by the model's position.

Ridge decision	Count	Mean y (bps)	SD (bps)	Positive (%)
Invested intervals	695	6.097	118.821	53.38
Cash intervals	307	2.121	107.137	53.75

The cash row records the SPY returns forgone while the portfolio held zero-interest cash. Values describe underlying asset returns before portfolio fees.

The same saved paths reconcile the net arithmetic return gap exactly. Write F_i^s for strategy s's gross return minus its net return, including any final liquidation charge. Since gross buy-and-hold earns y_i and gross Ridge earns $w_i y_i$,

$$252\big(\overline{R}^{\text{BH}} - \overline{R}^{\text{Ridge}}\big) = 252\,\overline{(1-w)y} + 252\big(\overline{F}^{\text{Ridge}} - \overline{F}^{\text{BH}}\big). \tag{16}$$

Table 11 gives the corresponding quantities. Missing positive equity returns costs 1.64 percentage points annually; incremental fees cost another 3.07 points. Together they account for the 4.71-point arithmetic shortfall. Fees explain about 65.20% of that gap, with reduced participation explaining the remainder. This is an additive return attribution; the 4.64-point CAGR gap additionally reflects compounding.

Table 11. Reconciliation of buy-and-hold's net arithmetic return advantage.

Component	Annual gap (pp)
Equity returns forgone during cash intervals	1.64
Incremental fee drag relative to buy-and-hold	3.07
Total net arithmetic return gap	4.71

Values are annual percentage points, calculated from the same 1,002 saved holding intervals at 5 bps. Fee drag is measured in realized interval returns.

5.4 Risk, drawdown, and market participation

Ridge's 15.72% annualized volatility is below buy-and-hold's 18.31%, and maximum drawdown is -22.20% rather than -26.28%. MA20 has lower volatility and drawdown still, at 10.83% and −19.14%, but earns the lowest CAGR. Figure 5 makes the growth–risk trade-off visible over time.

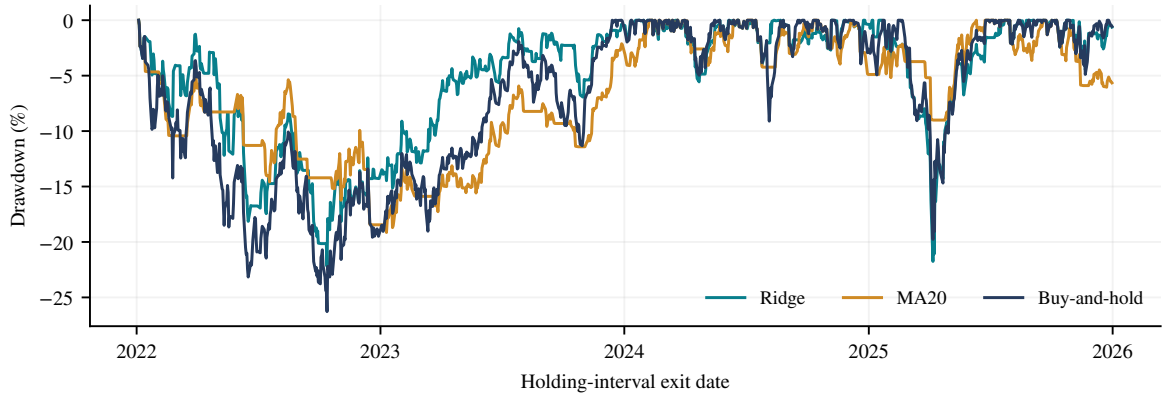


Figure 5. Net drawdowns from the running wealth peak, including the initial investment.

Average market participation is 69.36% for Ridge and 65.27% for MA20. Figure 6 shows the variation in monthly participation. These are averages of binary daily positions: a month with 70% exposure means the strategy was fully invested during roughly 70% of its holding intervals, not that it maintained a daily fractional position.

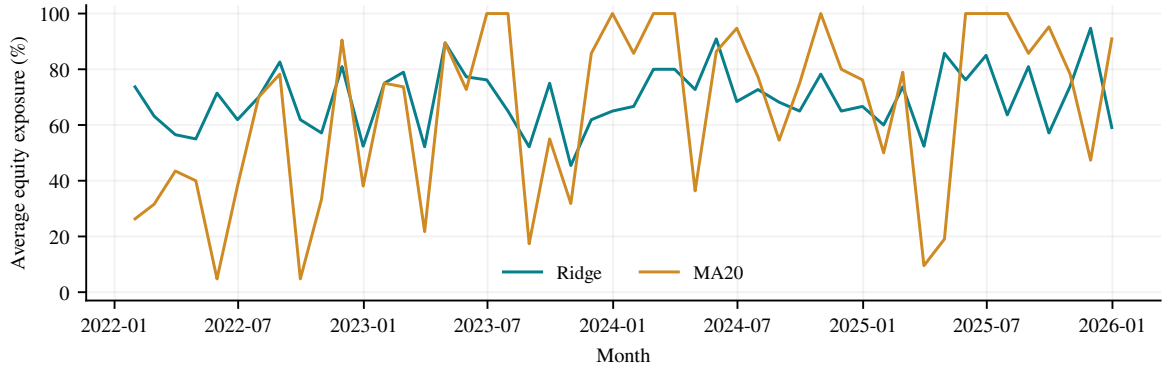


Figure 6. Monthly mean equity exposure for the two timing rules. Buy-and-hold remains fully invested throughout.

The saved ex post exposure diagnostic places a constant gross fraction of 69.36% in SPY. It has 8.03% CAGR, 12.70% volatility, and 0.672 Sharpe. For a fixed positive fraction p and zero cash return, multiplying every asset return by p scales its arithmetic mean and standard deviation equally, leaving gross Sharpe unchanged. This explains how lower equity participation alone can lower volatility. Ridge's observed risk reduction should therefore be evaluated alongside the lower net Sharpe, rather than treated as a separate timing success. The diagnostic's construction is discussed in Section 6.2.

Binary timing does not reduce each day's market position proportionally. It retains the full return during invested intervals and replaces it with zero during cash intervals. The underlying return standard deviation is 118.82 bps in the invested subset and 107.14 bps in the cash subset. Thus the observed allocation does not concentrate its equity exposure in the lower-volatility subset. Consistently, gross Ridge volatility is 15.71%, above the constant-fraction diagnostic's 12.70% despite matching its average exposure. The lower full-sample risk relative to a fully invested portfolio is closely connected to time spent in cash; the conditional return figures supply no corresponding evidence of successful low-risk interval selection.

5.5 Annual performance and rolling coefficients

Figure 7 presents annual net returns and Sharpe ratios. Ridge loses 13.80% in 2022, compared with 18.82% for buy-and-hold. It then earns 16.15%, 16.90%, and 9.88% in 2023–2025, below the corre-

sponding passive returns of 27.09%, 25.40%, and 17.79%. The reduction in the first year's loss does not compensate for the subsequent participation gap. The complete annual metrics are given in Appendix B.

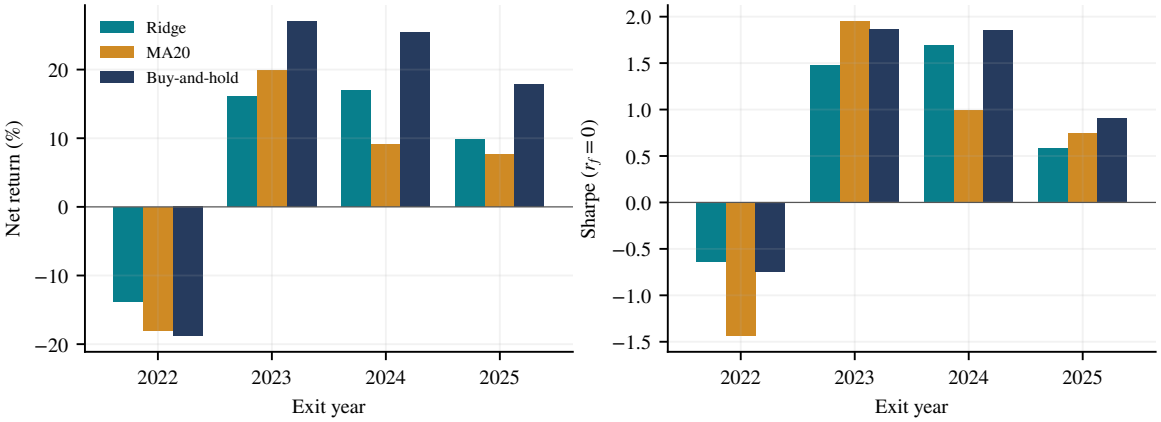


Figure 7. Calendar-year net returns and Sharpe ratios at 5 bps. Annual returns compound the continuous portfolio path without year-end liquidation.

Annual risk also varies. Ridge volatility is about 20.11% in 2022, 10.60% in 2023, 9.47% in 2024, and 19.59% in 2025. The 2025 within-year maximum drawdown is 21.75%, versus 19.76% for buy-and-hold. Thus the smaller full-period drawdown does not describe a uniform year-by-year reduction in losses. Average exposure remains between 65.60% and 73.02%, so the strategy continues to hold substantial equity risk in each year.

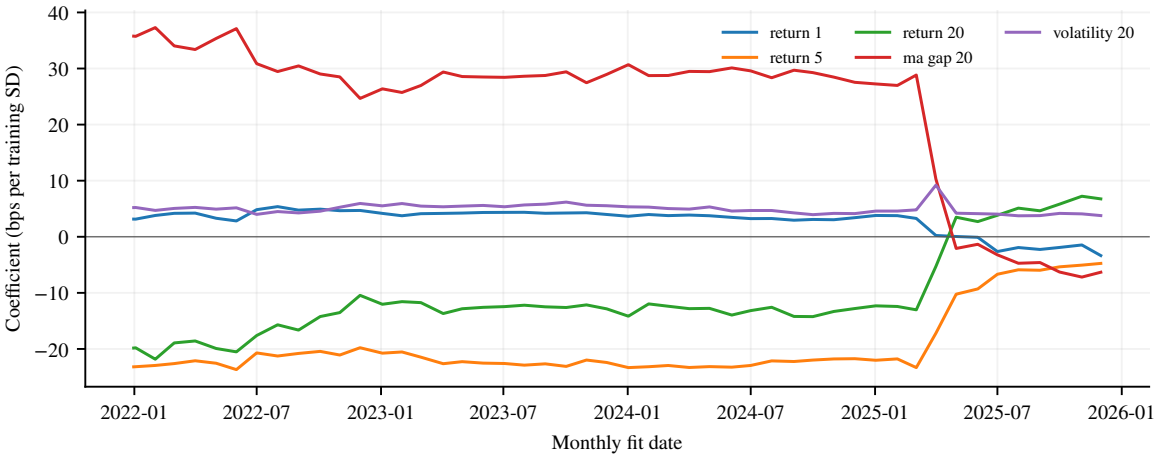


Figure 8. Monthly Ridge coefficients on training-standardized features, in bps per training standard deviation. Each curve contains the 49 recorded final-period fits.

Figure 8 relates the daily decisions to the fitted predictive relationship. The five-session return coefficient is negative in all 49 fits, while the volatility coefficient is positive throughout. Their mean standardized coefficients are −19.59 and 4.96 bps. The moving-average gap has a mean coefficient of 23.81 bps and changes sign once. Several coefficient paths change substantially around the second quarter of 2025, showing that the scheduled rolling procedure can alter its estimated relationship even with a fixed penalty and feature set.

Because the model combines several overlapping trend measures, the investment decision depends on their sum and the intercept. For example, the positive moving-average coefficient and negative five-session coefficient can offset each other. Monthly parameter changes are therefore examined together with the prediction and exposure paths, rather than by assigning each feature its own standalone trading rule.

6 Discussion and Limitations

6.1 What the experiment establishes

The main empirical result is a net performance shortfall relative to SPY buy-and-hold. Three measured components explain it. First, squared-error prediction is slightly worse than the historical-mean reference, and direction accuracy does not exceed the unconditional positive-return rate. Second, the strategy omits intervals with positive average equity returns, lowering gross growth. Third, 246 one-way trade units create a substantial fee burden. These components are consistent with the observed combination of lower volatility, lower CAGR, and lower net Sharpe.

The financial and statistical comparisons answer different questions. The negative R^2_{OOS} concerns squared error; the portfolio result concerns weighted returns and costs. Here both offer little support for the selected rule, but the connection is empirical rather than an identity. The strategy also selects an invested subset with a higher mean asset return than the excluded subset. That association is insufficient to offset foregone positive exposure and implementation costs. The joint evidence is more informative than either a hit rate or a drawdown ranking alone.

Ridge has a clear role within the workflow: it constrains coefficient magnitudes in a correlated-input regression. The rolling window and monthly schedule determine how much history informs each decision. Validation then chooses among four implementations of that estimator. This separation keeps the interpretation precise. The experiment evaluates a regularized trading procedure, not a theorem that shrinkage or more features must produce an investment advantage.

6.2 Limitations

The strategy was designed in November 2026 using already historical market data. The implemented configuration and selection were fixed before its initial download, but the study is not an externally preregistered prospective experiment. Selecting the largest validation Sharpe can favor noise even with four candidates. The fall from validation Sharpe 1.206 to test Sharpe 0.481 can reflect selection optimism, changing relationships, or sampling variation; the present design does not distinguish those explanations.

The final evaluation concerns one ETF and four years. Sharpe ratios and annual results are point estimates: no dependence-robust confidence intervals, formal superiority tests, or factor-alpha estimates are included. The gross Sharpe difference of approximately 0.007 is consequently treated as numerical similarity, not evidence of outperformance. Conditional-return and coefficient summaries are descriptive associations among correlated observations and inputs.

Execution assumes the observed next opening price and a fixed proportional fee. Opening-auction depth, market impact, partial fills, and actual brokerage financing are not modeled. Cash earns zero, so the reported Sharpe is a zero-rate measure rather than realized excess performance over Treasury bills. Dividend receivables are credited and reinvested at the ex-date exit open instead of their payment date. The supported history has no splits. Vendor revisions can alter a later download, and exact numerical reproduction depends on the recorded input snapshot.

The constant-exposure diagnostic uses the observed test-average allocation and ignores fractional rebalancing costs. It is an ex post attribution calculation, not an investable benchmark. A risk-matched strategy would require its own advance allocation rule and accounting. Likewise, no-trade regions, continuous weights, state interactions, other training windows, and additional assets are outside the evaluated strategy. These are changes to the research design rather than results of this study.

7 Conclusion

The selected Ridge rule converts five trailing price features into daily SPY or cash positions under explicit timing, training, and cost conventions. In 2022–2025 it earns 6.54% net CAGR and a 0.481 Sharpe ratio, below buy-and-hold's 11.18% and 0.670. Its smaller full-period drawdown accompanies lower equity participation. Weak forecast performance, positive returns during excluded intervals, and frequent full-position switches jointly explain the economic result. The project demonstrates the complete path from a plausible signal specification to a validated, reproducible trading evaluation, while preserving the finding that this implementation does not improve net performance.

8 Reproducibility and Contributions

The research implementation, configuration, and published result files are available in the accompanying repository [9]. Training and accounting are implemented in `project2/core.py`; data checks in `project2/data.py`; validation and final evaluation in `project2/pipeline.py`. The saved CSV and JSON outputs produce the report's tables and figures. Appendix C provides platform-specific environment activation, platform-independent analysis commands, package versions, and the artifact map.

The recorded clean-environment rerun used the same input snapshot in a fresh working directory and reproduced the validation grid, test forecasts, fit audits, and headline metrics byte for byte. Eleven integrity checks cover label maturity, future-data perturbations, scaling, execution and distribution alignment, costs, data validity, and drawdown. Full fit records and individual test cases remain in the code package; the report retains representative records and the complete performance comparisons.

Author contributions

Member	Contribution
Li Chongkai	Research design and economic motivation; data collection, cleaning, and quality checks; feature and label construction; information timing and leakage review.
Zhang Leyao	Ridge specification and regularization; training-only standardization and rolling estimation; historical-mean baseline; chronological validation and hyperparameter selection.
Zhang Yikai	Trading rules and benchmark implementation; signal-to-position mapping; execution, portfolio accounting, and costs; backtesting, integrity checks, and reproducibility.
Ye Tiantian	Out-of-sample prediction and portfolio evaluation; return, risk, and drawdown analysis; benchmark and cost comparisons; empirical figures, interpretation, and conclusions.

AI assistance

GPT-6.1 Sol (OpenAI) was used for code review and debugging, and for report organization and language editing. The authors verified the reported results and take responsibility for the analysis and submitted work.

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A Supporting Data and Timing Checks

A.1 Corporate actions and price adjustments

The execution target uses raw opening prices and an explicit distribution receivable, as in Eq. (1). The feature series uses adjusted close. Table A.1 compares vendor adjusted-close returns with explicit close-to-close cash-dividend returns at representative dates. Their sample mean absolute difference is 0.00594 bps, and the maximum absolute difference is 2.52010 bps. The differences arise under the source's adjustment convention; execution consistently follows the explicit-dividend target.

Table A.1. Representative dividend checks from the recorded snapshot.

Ex-date	Dividend (USD)	Vendor return (%)	Cash return (%)	Δ (bps)
2010-03-19	0.4800	-0.50618	-0.50410	-0.20801
2010-06-18	0.5310	0.10842	0.10790	0.05112
2025-09-19	1.8310	0.49528	0.49392	0.13685
2025-12-19	1.9930	0.90636	0.90367	0.26878

Vendor and explicit cash returns are percentages; Δ is in basis points. Source: `output/tables/dividend_event_checks.csv`.

For feature invariance, suppose a later corporate-action adjustment multiplies every price A_s, $s = t-20, \ldots, t$, by a common positive factor λ. Then

$$\frac{\lambda A_t}{\lambda A_{t-k}} - 1 = \frac{A_t}{A_{t-k}} - 1, \qquad \frac{\lambda A_t}{\frac{1}{20}\sum_{j=0}^{19} \lambda A_{t-j}} - 1 = \frac{A_t}{\frac{1}{20}\sum_{j=0}^{19} A_{t-j}} - 1. \tag{A.1}$$

The twenty one-session returns used by the volatility feature are unchanged for the same reason. Thus all five inputs are invariant to a common rescaling across the required lookback. A dedicated test checks this property numerically.

A.2 Representative fit boundaries

Table A.2 shows the beginning and end of the monthly fit audit. All 49 fits contain 1,260 mature outcomes. The maximum label exit is no later than the fit session. The full table and training means, scales, intercepts, and coefficients are retained in `output/tables/test_fit_audit.csv`.

Table A.2. Selected records from the monthly fit audit.

Fit date	First training signal	Last training signal	Latest label exit	n
2021-12-31	2016-12-28	2021-12-29	2021-12-31	1260
2022-01-03	2016-12-29	2021-12-30	2022-01-03	1260
2022-02-01	2017-01-30	2022-01-28	2022-02-01	1260
2025-10-01	2020-09-23	2025-09-29	2025-10-01	1260
2025-11-03	2020-10-26	2025-10-30	2025-11-03	1260
2025-12-01	2020-11-20	2025-11-26	2025-12-01	1260

The scaler uses population variance, $s_{d,j}^2 = |\mathcal{T}_d|^{-1} \sum_{t \in \mathcal{T}_d} (x_{t,j} - \mu_{d,j})^2$, matching `StandardScaler`. The volatility predictor itself uses the sample variance of twenty returns. Those are separate definitions. The eligible-label filter is applied before either scaling or model fitting.

B Complete Performance Tables

Table B.1. Cost sensitivity for all three fixed position paths.

Strategy	bps	CAGR (%)	Sharpe	MDD (%)	Final wealth
Ridge	0	9.88	0.678	-20.97	1.4546
MA20	0	5.03	0.508	-17.84	1.2156
Buy-and-hold	0	11.20	0.672	-26.28	1.5254
Ridge	5	6.54	0.481	-22.20	1.2863
MA20	5	3.69	0.389	-19.14	1.1552
Buy-and-hold	5	11.18	0.670	-26.28	1.5238
Ridge	10	3.29	0.284	-23.89	1.1373
MA20	10	2.37	0.271	-20.43	1.0977
Buy-and-hold	10	11.15	0.669	-26.28	1.5223

The single-side fee is stated in bps. Final wealth includes initial entry and terminal liquidation.

Table B.2. Annual net return, risk, and participation at 5 bps.

Strategy	Year	Return (%)	Vol. (%)	Sharpe	Exposure (%)	Trades
Ridge	2022	-13.80	20.11	-0.644	65.60	60
Ridge	2023	16.15	10.60	1.478	67.20	69
Ridge	2024	16.90	9.47	1.696	73.02	59
Ridge	2025	9.88	19.59	0.582	71.60	58
MA20	2022	-17.99	13.28	-1.439	42.00	32
MA20	2023	19.93	9.61	1.954	67.20	21
MA20	2024	9.09	9.18	0.993	80.16	23
MA20	2025	7.67	10.69	0.751	71.60	26
Buy-and-hold	2022	-18.82	24.15	-0.750	100.00	1
Buy-and-hold	2023	27.09	13.47	1.861	100.00	0
Buy-and-hold	2024	25.40	12.65	1.854	100.00	0
Buy-and-hold	2025	17.79	20.41	0.910	100.00	1

Return is compounded realized annual return; volatility and Sharpe use 252 intervals. Exposure is the average binary equity position. The portfolio is continuous across year boundaries.

C Reproducing the Analysis and Report

C.1 Environment and analysis commands

Clone the research repository and select the analysis version:

```
git clone https://github.com/SpencerMuZi/DSA5205-p2.git
cd DSA5205-p2
git checkout 2e2e265ba9d007c94279225febaa15a054dff7cf
```

Use Python 3.12. Ridge uses deterministic SVD; seed 42 controls synthetic test fixtures. From the repository root, create an environment with `python -m venv .venv` and activate it:

```
# Windows PowerShell
.venv\Scripts\Activate.ps1
# macOS / Linux
source .venv/bin/activate
# Then, on either platform:
python -m pip install -r requirements-lock.txt
python -m pytest -q
python -m project2.pipeline all --workspace runs/reproduction
```

The pipeline writes its data audit, validation selection, forecasts, and performance results to `runs/reproduction/output/`. The published snapshot has 4,024 sessions and selects $\alpha = 1$; Table 6 gives the reference performance.

Table C.1. Software versions recorded for the published research run.

Component	Version	Component	Version
Python	3.12.14	NumPy	2.3.5
pandas	2.2.3	scikit-learn	1.9.1
Matplotlib	3.11.2	yfinance	1.8.0
exchange-calendars	4.13.2	pytest	9.2.0

The complete dependency specification is in `requirements-lock.txt`; the original run metadata are in `output/run_manifest.json`.

C.2 Source-to-result correspondence

Table C.2. Analysis entry points and principal saved artifacts. Paths are repository-relative.

Source or artifact	Role
`config.json`	Asset, periods, window, grid, costs, and position rule
`project2/data.py`	Download, exchange-calendar audit, and corporate actions
`project2/core.py`	Features, mature labels, forecasts, accounting, and metrics
`project2/pipeline.py`	Validation selection, fingerprints, and final evaluation
`tests/test_integrity.py`	Eleven timing, leakage, accounting, and data checks
`output/selection.json`	Selected penalty and experiment fingerprints
`output/tables/test_predictions.csv`	Decision, entry, exit, forecast, realized return, and position
`output/tables/test_fit_audit.csv`	All fitted-window boundaries, scalers, and coefficients
`output/tables/test_path_*.csv`	Per-interval returns, trades, fees, and wealth
`output/reproducibility_check.json`	Recorded clean-environment comparison

Input and output identities are recorded in `data/provenance.json` and `output/run_manifest.json`; manuscript source identities accompany the report archive.

Figure 1 uses `config.json`; Figures 2, 5, and 6 use `test_path_*.csv`. Figures 3, 4, 7, and 8 use `cost_sensitivity.csv`, `test_predictions.csv`, `annual_metrics.csv`, and `test_fit_audit.csv`, respectively. All CSV files are in `output/tables/`.

Compile the self-contained report archive using `latexmk -xelatex main.tex` (XeLaTeX and BibTeX). Its tables and vector figures require no data download.